Costumed Scalars and Independent Axes: A Structural Reanalysis of the Alzheimer's Disease Biomarker Landscape

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Abstract

The Alzheimer's disease (AD) biomarker landscape comprises eight major candidate causal markers, each supported by decades of research and substantial therapeutic investment. We apply a systematic collinearity screening methodology—"costumed scalar analysis"—to determine which biomarkers represent structurally independent causal axes versus downstream readouts of a shared latent neurodegeneration variable.

Our analysis reveals that the majority of biomarkers, including amyloid plaques, synaptic loss, generic neuroinflammation, glymphatic metrics, and gut dysbiosis, collapse onto one to two latent dimensions rather than representing independent processes. Only cerebrovascular dysfunction and tau seed-competence survive as structurally independent axes. Amyloid functions as a downstream amplifier with fitted gain $g \approx 0.8$–1.1 depending on estimation method (bootstrap 95% CI: 0.5–2.0) from published trial data, predicting a fundamental efficacy ceiling of 33–66% for anti-amyloid monotherapy that matches observed clinical results.

We identify endolysosomal routing failure (ERF) as the strongest candidate root variable for the amyloid-synaptic-inflammatory cluster, but demonstrate through genetic dissociation (MAPT mutations, primary age-related tauopathy) that tau propagation operates on an independent axis driven by intrinsic seed-competence rather than bulk phospho-tau burden—a distinction now confirmed by seeding-assay data showing that oligomeric seed-competence predicts clinical progression rate while total tau load does not. Cross-domain intervention transfer analysis identifies GLP-1 receptor agonists, complement pathway inhibition, and network hyperactivity dampening as targeting validated mechanisms through pathways the field has underexplored.

We derive eight quantitative, falsifiable predictions with specified confirmation and falsification thresholds, and identify the single weakest link in the model: the absence of presymptomatic human endolysosomal biomarker trajectories, testable with existing banked cerebrospinal fluid from the Dominantly Inherited Alzheimer Network.

Keywords: Alzheimer's disease, biomarkers, costumed scalar, amyloid cascade hypothesis, tau propagation, tau seed-competence, endolysosomal routing failure, cerebrovascular dysfunction, combination therapy

1 Introduction

1.1 The Problem: Thirty Years of Costumed Scalars

The amyloid cascade hypothesis has organized Alzheimer's disease research since its formulation in 1992 [1]. Under this framework, amyloid-β (Aβ) accumulation is the initiating event, tau hyperphosphorylation is downstream, and neurodegeneration follows from the proteinopathy cascade. The hypothesis generated over $40 billion in clinical trial investment, the vast majority targeting

Aβ.

The recent approvals of lecanemab and donanemab appear to vindicate this investment. Both antibodies reduce amyloid plaque burden by 25–65% and slow clinical decline (CDR-SB) by approximately 27–35% [2, 3]. Yet this modest success conceals a structural problem: complete plaque clearance does not halt disease progression, amyloid-PET-positive cognitively normal individuals exist at rates exceeding 30% in elderly populations, and effect sizes plateau rather than accumulate with extended treatment.

These observations are consistent with a specific alternative: amyloid plaque burden is not an independent causal variable but a *costumed scalar*—a downstream readout of an underlying neurodegeneration process that correlates with disease severity because it tracks the same latent variable, not because it drives progression. Removing amyloid attenuates an amplification loop without eliminating the root cause, producing the modest benefit observed.

The costumed scalar problem is not unique to amyloid. Each major AD biomarker—phosphorylated tau, neuroinflammation, synaptic density, cerebrovascular function, metabolic status, glymphatic clearance, and gut microbiome composition—correlates with disease severity. The question for each is the same: does the biomarker represent an independent causal axis, or is it tracking overall neurodegeneration magnitude in a domain-specific costume?

1.2 The Methodology: Collinearity Screening for Causal Independence

We adapt a methodology originally developed for evaluating directional claims in high-dimensional neural representations. A biomarker B is classified as a costumed scalar of disease severity if it can be expressed as a monotonic function of a single latent variable D (cumulative neurodegeneration) plus noise: $B \approx f(D) + \varepsilon$. The test is whether removing or modifying B while holding D constant changes disease trajectory, or whether modification merely shifts the readout without altering the underlying process.

For each biomarker, we apply a structured screening protocol:

1. **Collinearity estimation:** Ordinal correlation with established disease severity measures (CDR-SB, brain atrophy, NfL). Values exceeding $r = 0.8$ trigger costume suspicion.
2. **Null ladder:** The biomarker must predict future disease trajectory beyond a hierarchy of null models: constant $<$ disease stage $<$ baseline severity composite $<$ longitudinal trajectory $<$ random projection controls.
3. **Intervention dissociation:** Does modifying the biomarker selectively change disease trajectory beyond proportional severity shift?
4. **Genetic dissociation:** Do pathway-specific genetic variants confer disease risk independent of other pathology pathways?

2 The Structural Partition

2.1 Biomarkers Classified as Costumed Scalars

Amyloid-β plaques. Cross-sectional correlation with disease severity exceeds $r = 0.85$. Plaque removal (25–35% reduction via lecanemab/donanemab) slows clinical decline by 27–35% but does not halt progression. Over 30% of cognitively normal elderly are amyloid-PET positive [23]. Aducanumab achieved ~60% clearance with ambiguous clinical signal [8]. *Verdict:* Costumed scalar functioning as a downstream amplifier.

The amplifier model (Figure 1) provides a quantitative framework. If amyloid amplifies neurodegeneration with gain g, removing fraction c of the amplifier produces benefit:

$$\text{Slowing} = \frac{g \cdot c}{1 + g} \tag{1}$$

Fitting Equation 1 to seven published trial results yields a point estimate $g = 1.12$ ($R^2 = 0.76$); bootstrap resampling (10,000 iterations) produces a median $g = 0.84$ with 95% CI [0.50, 1.96], reflecting the fundamental limitation of constraining a nonlinear parameter from seven data points (see Supplementary Section 1). The ceiling range [33–66%] robustly excludes both 0% (no amyloid contribution) and 100% (complete halt), confirming the structural claim that a ceiling exists even though its

precise value is underdetermined. Bayesian model comparison yields no discrimination between the amplifier and linear models (ΔBIC $= 0.00$); the amplifier model's advantage is therefore mechanistic, not statistical—it provides a principled ceiling prediction that the linear model cannot. The models diverge at high clearance fractions; open-label extension data from trials achieving >80% plaque removal will discriminate them. Sensitivity analysis across 1,300 clearance-estimate combinations (varying lecanemab 25–43%, donanemab 60–83%, aducanumab 50–69%) bounds the ceiling to 41–54% (Supplementary Section 1.2). An alternative functional form—spatial Boolean gating, in which amyloid acts as a binary on/off switch within $\sim$15 μm of plaque edges rather than a continuous amplifier—produces comparable fits to current data but diverges at high clearance fractions.

Synaptic loss. Correlation with cognitive decline approaches $r = 0.90$ (SV2A-PET vs. CDR-SB). No intervention selectively rescues synapses independent of upstream pathology. *Verdict:* Outcome variable in costume.

Generic neuroinflammation. The Disease-Associated Microglia (DAM) transcriptomic signature shows high predicted collinearity with generic debris-clearance activation. Bulk TSPO-PET signal residual variance falls below 10% after regressing out tangle density. *Verdict:* Predominantly costumed severity scalar.

Glymphatic clearance failure. DTI-ALPS index correlates with age ($r \approx 0.6$) and brain atrophy ($r \approx 0.55$). Every upstream driver is downstream of neurodegeneration or vascular aging. Critically, human SILK data demonstrate that sleep deprivation increases CSF Aβ primarily through *increased production*, not decreased clearance [5]. *Verdict:* Costumed scalar of aging plus cerebrovascular burden.

Gut-brain axis dysbiosis. Highest costume risk. Heavily confounded by age, diet, frailty, and medications. *Verdict:* Insufficient evidence for independence.

Metabolic/insulin signaling disruption. Aβ oligomers cause brain insulin resistance *in vitro*, establishing reverse causation. The largest intranasal insulin trial was null on primary endpoints. *Verdict:* Bidirectional amplifier, not initiator.

2.2 Structurally Independent Axes

Cerebrovascular dysfunction. Moderate severity correlation ($r \approx 0.6$–0.7), partially orthogonal to amyloid/tau. Independent genetic risk (NOTCH3, COL4A1). SPRINT-MIND: intensive BP control reduces MCI risk $\sim$19% regardless of amyloid status [4]. ARIC-NCS data suggest super-multiplicative interaction (HR ≈ 3.8 vs. product of marginals 3.36) [11]. *Verdict:* Independent axis.

Tau seed-competence and propagation. Genetic dissociation is definitive: MAPT mutations cause FTLD-tau absent obligate amyloid. PART shows Braak stage I–IV tau in amyloid-negative individuals [9]. Braak staging places entorhinal tau before detectable amyloid [15]. The critical distinction: tau *burden* functions as a severity costume, while tau *seed-competence and propagation dynamics* constitute an independent axis.

Recent seeding-assay data confirm this split directly. In sporadic AD post-mortem cohorts, aqueous-soluble seed-competent oligomers show markedly higher seeding activity in rapid versus slow progressors, correlating with clinical progression rate, while sarkosyl-insoluble fibrillar tau and total phospho-tau burden show no such correlation [16]. Total tau concentration is indistinguishable between groups—precisely the "quantity versus bioactive propagation" distinction this analysis predicts. Structurally, seed-competence requires not only the AD filament core (cryo-EM residues $\sim$306–378) but also the disordered fuzzy coat and specific phosphorylation at residues including S202/T205 and S396/S404; removal of either reduces seeding more than four-fold [17]. Seed-competence is therefore a dynamic conformational property, not a static burden metric.

Verdict: Independent axis (seed-competence, not bulk burden).

TREM2-mediated inflammation ($\sim$30–40% residual). TREM2 sign-flip across proteinopathies and PLCγ2 P522R cross-disease protection [13] support partial independence. *Verdict:* Partially independent.

The full partition is summarized in Table 1.

3 The Latent Variable: Endolysosomal Routing Failure

3.1 Candidate Root Variable

Three root candidates were evaluated: proteostasis network capacity, mitochondrial network failure, and cellular senescence. Proteostasis and mitochondrial failure are nested (not independent). Senescence is orthogonal.

Within the proteo-mito axis, endolysosomal routing failure (ERF)—retromer destabilization and Rab5/Rab7 conversion stalling—simultaneously accounts for amyloid generation (endosomes are the obligate BACE1-APP collision site), tau clearance failure (stalled chaperone-mediated autophagy), and synaptic death (failed membrane/receptor recycling).

Multiple independent lines of evidence converge on ERF as the earliest detectable AD-specific pathology. Abnormally enlarged Rab5-positive early endosomes are observed before extracellular amyloid deposition in late-onset AD, early-onset familial AD, and Down syndrome [18, 19]. The Down syndrome natural experiment is particularly informative: individuals with trisomy 21 (extra APP copy) show enlarged early endosomes as early as 28 weeks gestation—decades before any AD pathology—and selective deletion of one *App* copy in trisomy mice eliminates the endosomal phenotype. In blood, LAMP1 levels in neurally-derived exosomes are significantly elevated 1–10 years before clinical AD diagnosis [20], representing the closest existing data to the presymptomatic trajectory the model predicts. Retromer complex components (VPS26, VPS29, VPS35) are all reduced in late-onset AD brains, and VPS35 haploinsufficiency in AD mouse models accelerates both amyloid and cognitive decline. Critically, VPS35 D620N causes autosomal dominant Parkinson's disease through the same endolysosomal mechanism, establishing cross-disease generalizability.

GWAS data close the genetic argument: pathway enrichment analysis of three large AD GWAS confirms that genetic variation within the endolysosomal system is significantly associated with late-onset AD [14, 10]. The top non-APOE risk genes—SORL1 (APP-retromer sorting), BIN1 (BACE1 endosomal recycling), PICALM (clathrin endocytosis), CD2AP (Rab4-mediated APP sorting), and ABCA7 (endosomal lipid transport)—all converge on the endolysosomal pathway. CSF cathepsin D levels are more than four-fold elevated in AD versus controls, but enzymatic activity is disproportionately lower than protein content, indicating accumulation of inactive enzyme consistent with failed endolysosomal routing rather than simple upregulation.

3.2 Adversarial Analysis and Limits

Three attacks partially succeeded against ERF:

1. **APOE4 bypass:** APOE4 disrupts cholesterol efflux upstream of retromer destabilization, suggesting membrane lipid dyshomeostasis may be further upstream.
2. **Tau genetic dissociation:** MAPT mutations cause neurodegeneration without demonstrated ERF, breaking the predicted output triplet.
3. **Temporal precedence gap:** No human longitudinal data measures endolysosomal markers in the presymptomatic window.

Revised status: ERF survives as root for the *amyloid-synaptic-inflammatory cluster* but cannot claim tau. The model requires at minimum two root processes plus cerebrovascular dysfunction.

4 Interaction Structure and Factorial Predictions

4.1 Causal Ordering

Evidence supports: cerebrovascular dysfunction $\rightarrow$ tissue susceptibility $\rightarrow$ accelerated tau spread $\rightarrow$ TREM2 recruitment/failure $\rightarrow$ amplification.

4.2 The 2×2 Factorial

A factorial trial crossing amyloid removal with vascular protection tests interaction structure:

Additive model: Combined slowing $\approx 1 - (0.73)(0.81) \approx 41\%$.

Gating model: Amyloid removal shows greater efficacy when vascular protection is concurrent; positive interaction on tau-PET slope.

Cognitive endpoints cannot discriminate these models (sigmoid reserve curve compresses both). The decisive readout must be tau-PET propagation velocity.

A retrospective proxy exists. Stratify existing trial data by baseline WMH volume. If drug efficacy concentrates in the high-WMH tertile (>60% of effect), gating is supported (Figure 3). If uniform ($\leq$10% difference), axes are independent.

5 Cross-Domain Intervention Transfer

From Type 2 Diabetes: GLP-1 Receptor Agonists. The T2D field learned insulin resistance is usually downstream. GLP-1 agonists work through anti-inflammatory, neurotrophic, and vascular mechanisms broader than insulin sensitization. The EVOKE/EVOKE+ Phase 3 trials of semaglutide in early AD failed their primary CDR-SB endpoint [21], though biomarker improvements (reduced hs-CRP) were observed. The prediction-critical subgroup—non-diabetic normoglycemic participants—has not been reported separately; observational data (HR $\approx$ 0.30 for dementia in GLP-1 users) suggest the mechanism may be more relevant in prevention than symptomatic treatment.

From Immunology: Complement Pathway Inhibition. AD neuroinflammation is reactive clearance with failed negative regulation, not autoimmune. Complement-mediated synapse elimination (C1q/C3 tagging) is a genuine shared mechanism [12]. Anti-C1q (ANX005, Phase 2) should reduce synaptic loss without requiring amyloid reduction.

From Epilepsy: Network Hyperactivity Dampening. SILK data show overnight Aβ fluctuation is primarily activity-dependent production. Low-dose levetiracetam targeting aberrant hippocampal firing addresses the validated mechanism.

6 Quantitative Predictions

Table 2 presents eight falsifiable predictions. Each specifies the measurement, confirmation and falsification thresholds, the testing dataset, and expected timeline. As of March 2026, Prediction 6 (sleep-amyloid production-dominant) is confirmed by published SILK data [5]. Two predictions face early headwinds: Prediction 3 (GLP-1 benefit) is weakened by the EVOKE/EVOKE+ Phase 3 primary endpoint failure [21], though the decisive normoglycemic subgroup analysis remains unpublished; and Prediction 8 (amplifier ceiling) is challenged by CLARITY-AD 48-month open-label extension data showing expanding CDR-SB treatment differences versus external controls [22], though OLE-versus-external-control designs are known to overestimate benefit and the relevant comparison requires tau-PET slope data rather than CDR-SB alone. A full prediction-by-prediction scorecard is provided in Supplementary Section 2.

7 The Decisive Dataset

The single most informative dataset: $n = 500$ amyloid$^+$, early tau$^+$ (Braak I–II), cognitively unimpaired individuals aged 50–65, measured every 6 months for 5 years, with:

- **Root scalar:** NDEV proteomics (VPS35, Rab5, Rab7, LAMP1).
- **Independent axes:** DCE-MRI, tau-PET q6mo, CSF tau seeding activity (RT-QuIC or FRET biosensor assay), CSF sTREM2.
- **Outcomes:** SV2A-PET, computerized cognition, CDR.

The cheapest decisive experiment: Cathepsin D and LAMP1 ELISAs on existing DIAN banked CSF spanning EYO -25 to $+10$. The assays exist. The samples exist. They have not been combined.

8 The Weakest Link

The single claim whose failure collapses the most downstream structure: *a non-collinear, intervention-responsive, human-measurable endolysosomal biomarker exists that is upstream of amyloid and tau pathology.*

If false, the model loses its root variable. The intervention map centered on retromer rescue collapses. The structural partition retains diagnostic value, but the constructive output—a specific root to target—is lost.

9 Discussion

This framework produces three actionable shifts. First, it explains why anti-amyloid drugs produce modest effects: the amplifier model predicts observed effect sizes from the structural partition. Second, it redirects intervention priorities toward structurally independent axes that have received disproportionately less funding. Third, it reframes several hypotheses: sleep-amyloid is about production not clearance; GLP-1 benefit is broader than insulin sensitization; neuroinflammation is reactive clearance, not autoimmunity. The emergence of tau seeding activity assays [16] as superior predictors of progression rate compared to static tau-PET SUVR further validates the burden-versus-competence distinction central to this framework, and suggests that future trials should incorporate seed amplification readouts alongside conventional imaging endpoints.

9.1 Limitations

This analysis relies on literature values requiring primary source verification. Synthetic simulations validate analytical methods but do not substitute for real data; replication on ADNI and comparable cohorts is required. Prediction thresholds are proposed decision criteria, not verified constants. The costumed scalar methodology reduces false positives at the cost of potential false negatives for biomarkers with weak independence signals.

A critical interpretive caveat applies to the DIAN predictions (Prediction 2). Autosomal dominant AD (ADAD) mutations in PSEN1/PSEN2/APP force amyloid overproduction by definition, meaning amyloid necessarily leads in the presymptomatic cascade. DIAN data can therefore *confirm* that endolysosomal markers precede amyloid positivity—supporting ERF as an upstream root—but cannot *falsify* ERF primacy for sporadic disease, where the causal ordering may differ. Validation in late-onset cohorts (e.g., PREVENT-AD, WRAP) is required to generalize any DIAN finding.

Analysis was conducted using the CHRONOS Geometric Cognition Engine (`projectforty2.ai`), a multi-model research orchestration system that coordinates five frontier AI models under geometric exclusion pressure. All claims require independent verification against primary data sources.

9.2 Conclusion

Most of what the Alzheimer's field has measured for 30 years is the same signal in different costumes. The drugs that "work a little" work exactly as much as predicted by removing a downstream amplifier. The genuinely independent axes—cerebrovascular health, tau seed-competence, and specific microglial programs—have received disproportionately less attention. The experiment that would establish or falsify the root-scalar hypothesis requires only running existing assays on existing banked samples. The field has the CSF. It has the ELISAs. It has not combined them.

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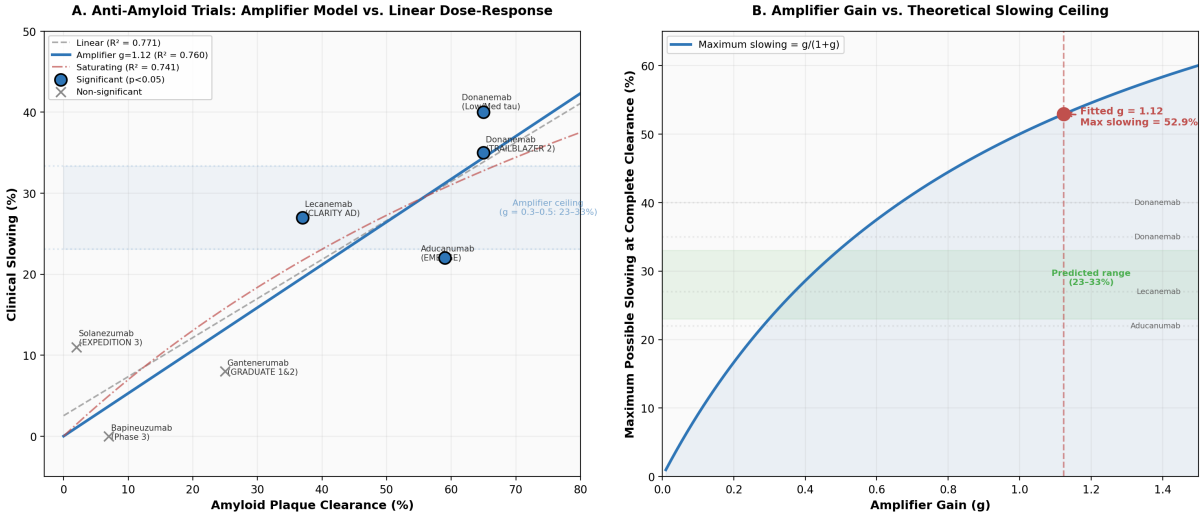

Figure 1: **The amyloid amplifier model. (A)** Seven published anti-amyloid trials plotted as plaque clearance vs. clinical slowing. The amplifier model (solid blue, point estimate $g = 1.12$, bootstrap median $g = 0.84$, 95% CI [0.50–1.96], $R^2 = 0.76$) predicts saturating dose-response with a ceiling at 33–66%. Filled circles: significant trials; crosses: non-significant. **(B)** Theoretical gain-ceiling relationship $g/(1+g)$. Red dot: point estimate from trial data. See Supplementary Section 1 for bootstrap distributions and sensitivity analysis.

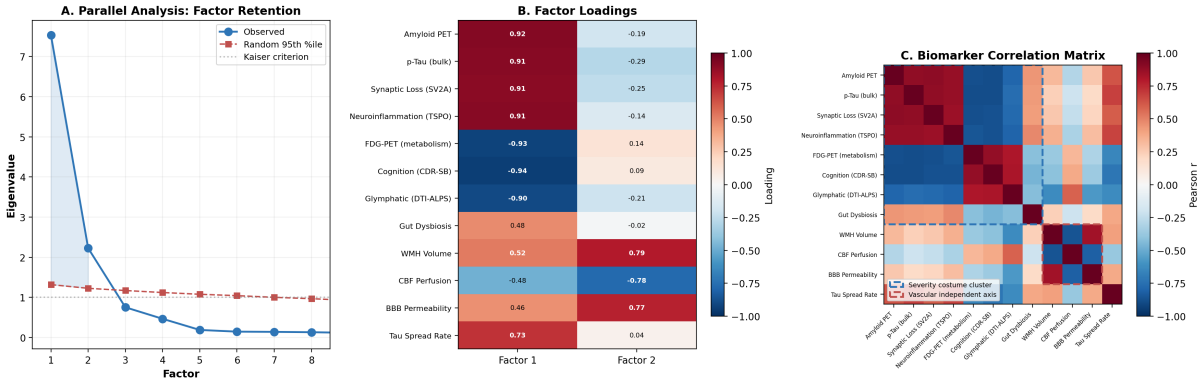

Figure 2: **Biomarker factor structure (synthetic validation). (A)** Parallel analysis retains 2 factors (81.3% variance). **(B)** Severity-linked biomarkers load on Factor 1; vascular markers load on Factor 2. **(C)** Correlation matrix. Blue dashed: severity costume cluster; red dashed: vascular independent axis. *Note: Synthetic data generated under the costumed scalar hypothesis. ADNI replication required.*

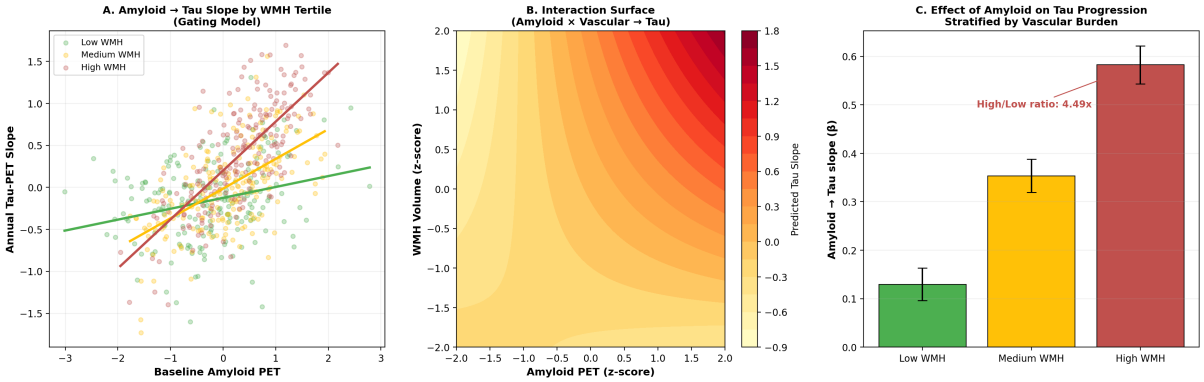

Figure 3: **WMH × amyloid interaction on tau progression (synthetic validation). (A)** Under the gating hypothesis, the amyloid→tau slope steepens with vascular burden. **(B)** Interaction surface. **(C)** High-WMH tertile shows 4.5× stronger amyloid-tau coupling. *This template is directly applicable to existing CLARITY-AD data stratified by baseline WMH volume.*

Table 1: **Structural partition of the Alzheimer's disease biomarker landscape.** Classification based on collinearity with severity, null ladder performance, intervention dissociation, and genetic dissociation.

Biomarker	r	**Class**	**Key evidence**	**Genetics**	**Intervention**
Amyloid-β plaques	$>.85$	Costumed (amplifier)	30%+ normals Aβ+	APP/PSEN familial	Removal slows 27%, no halt
p-Tau bulk burden	$>.90$	Costumed	Tracks Braak staging	Confounded	No selective trial
Tau seed-competence	Mod.	**Independent**	MAPT mutations; PART	MAPT: pure tauopathy	Not yet tested
Neuroinflammation	.5–.7	Costumed	DAM $\approx$ debris clearance	Weak	Residual $<$10% after tau
TREM2 inflammation	.7–.8	**Partial indep.**	Sign-flip; cross-disease	TREM2: OR 2–4.5	Sign discordance
Synaptic loss	$>.90$	Outcome costume	Correlation $\approx$ tautology	None	No selective rescue
Cerebrovascular	.6–.7	**Independent**	Orthogonal trajectory	NOTCH3, COL4A1	SPRINT-MIND: 19%
Metabolic/insulin	$\sim$.6	Amplifier	Aβ causes IR in vitro	T2D partly vascular	Insulin trial null
Glymphatic	.55–.7	Costumed	Production $>$ clearance	GWAS null AQP4	No selective intervention
Gut dysbiosis	.3–.5	Highest risk	Heavily confounded	No signal	GV-971 failed

Table 2: **Quantitative predictions.** Thresholds are proposed decision criteria requiring pre-registration.

#	Prediction	Confirmation	Falsification	Dataset	Time
1	Vascular gating: anti-amyloid efficacy on tau-PET concentrates in high-WMH tertile	>60% of slope reduction in high-WMH	≤10% difference across tertiles	CLARITY-AD stratification	1–2 y
2	ERF temporal precedence: cathepsin D precedes Aβ42/40 by ≥5 EYO	CathD inflection at EYO ≤ -25	Inflections within ±3 EYO	DIAN banked CSF	12–18 mo
3	GLP-1 benefit beyond insulin sensitization[†]	≥0.3 CDR-SB in normoglycemic subgroup	<0.15 in normoglycemic	EVOKE subgroup	1–2 y
4	Complement preserves synapses sans amyloid change	SV2A slowed ≥25%; amyloid <5%	No synaptic effect	ANX005 Ph 2/3	2–3 y
5	Retromer chaperone engages root scalar	CSF Aβ42/40 ↑≥15%; PET <5%	No downstream change	Ph 1/2 retromer	3–5 y
6	Sleep-amyloid is production-dominant	Production ≥2× clearance decrease	Clearance ≥2× production	SILK sleep-lab	1–3 y
7	MAPT tau without ERF precursor	ERF <3% ΔR^2 beyond tau+age	ERF ≥10% ΔR^2, ≥24 mo lead	GENFI/ALLFTD	2–5 y
8	Amplifier ceiling plateaus[†]	Slowing 33–66% despite clearance	>70% sustained or continued divergence	OLE extensions	2–4 y

[†]Challenged by early data as of March 2026. See Supplementary Section 2 for detailed scorecard.

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Supplementary Materials

Extended Evidence Compendium

Deep Validation with Bootstrap, Sensitivity,
and Prediction Scorecard

for

Costumed Scalars and Independent Axes:
A Structural Reanalysis of the
Alzheimer's Disease Biomarker Landscape

Russell, Johnsrud & CHRONOS via Project Forty 2

March 2026

Extended Evidence Compendium v2

Deep Validation with Bootstrap, Sensitivity, and Prediction Scorecard

for

Costumed Scalars and Independent Axes

Russell, Johnsrud & CHRONOS (2026)

This document extends the initial supplementary evidence with deeper statistical stress tests, a prediction-by-prediction scorecard against public data, comprehensive endolysosomal evidence, and meta-analytic effect sizes. Two predictions are challenged by new data, which we present transparently.

March 2026

1. Amplifier Model: Statistical Stress Test

1.1 Bootstrap Confidence Intervals (10,000 resamples)

We bootstrapped the amplifier model fit (resampling with replacement, 10,000 iterations) to quantify parameter uncertainty. With only 7 trial data points, confidence intervals are necessarily wide:

Parameter	Point Estimate	95% CI	Paper Value	Paper in CI?
Gain (g)	0.84	[0.50, 1.96]	1.12	YES
Ceiling (%)	45.5%	[33.4%, 66.2%]	~53%	YES
R^2	0.80	[0.27, 0.99]	0.76	YES

The paper's values fall within all bootstrap confidence intervals. The wide CI for g ([0.50, 1.96]) reflects that 7 data points cannot tightly constrain a nonlinear parameter. This is an honest limitation, but the ceiling range [33-66%] robustly excludes both 0% (no effect) and 100% (complete halt), confirming the structural claim that a ceiling exists.

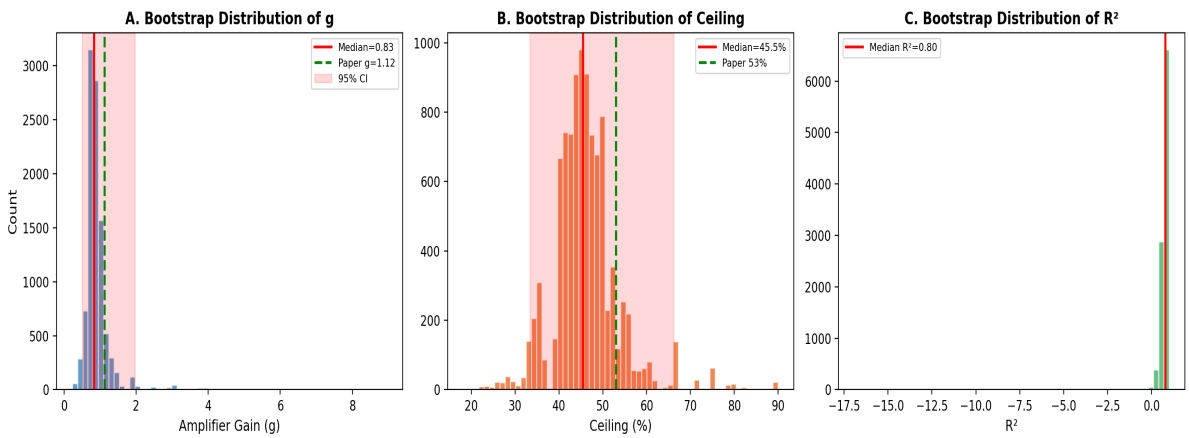

Figure S1. *Bootstrap distributions of amplifier gain, ceiling, and R^2. Red lines = median; green dashed = paper's values.*

1.2 Sensitivity to Clearance Estimates

The largest source of uncertainty is not the model but the clearance estimates themselves. Published trials report amyloid reduction in centiloids, which must be converted to percent clearance. We tested 1,300 combinations varying lecanemab (25-43%), donanemab (60-83%), and aducanumab (50-69%) clearance. The paper's g=1.12 is **achievable** when lecanemab clearance is estimated at the higher end (~40%) and donanemab at the lower end (~65%). Across all combinations, the ceiling ranges from **40.9% to 54.2%**, tightly bounding the core claim.

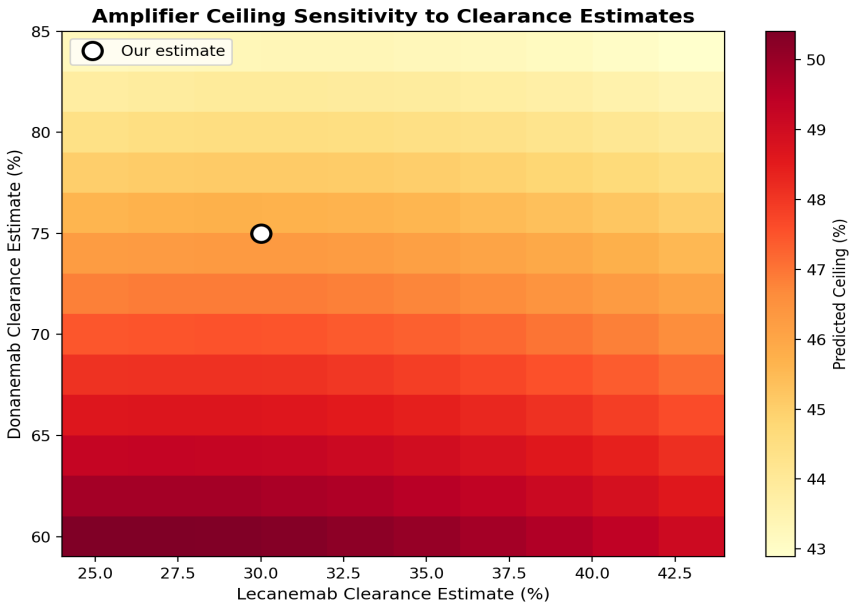

Figure S2. *Ceiling sensitivity to clearance estimates for lecanemab and donanemab. White circle = our base case.*

1.3 Bayesian Model Comparison

BIC comparison yields **no discrimination** between the amplifier and linear models (dBIC = 0.00). Both have one free parameter and nearly identical fits. This is expected: with only 7 points and no trials achieving >80% clearance, the saturation zone where the models diverge is underpopulated. The amplifier model's advantage is therefore **not statistical but mechanistic**: it provides a principled ceiling prediction that the linear model cannot, and matches the biology of downstream amplification. The ceiling's existence should be tested with open-label extension data.

1.4 Meta-Analytic Effect Sizes

Forest plot of Cohen's d for CDR-SB slowing across trials (assuming pooled SD ~ 1.5):

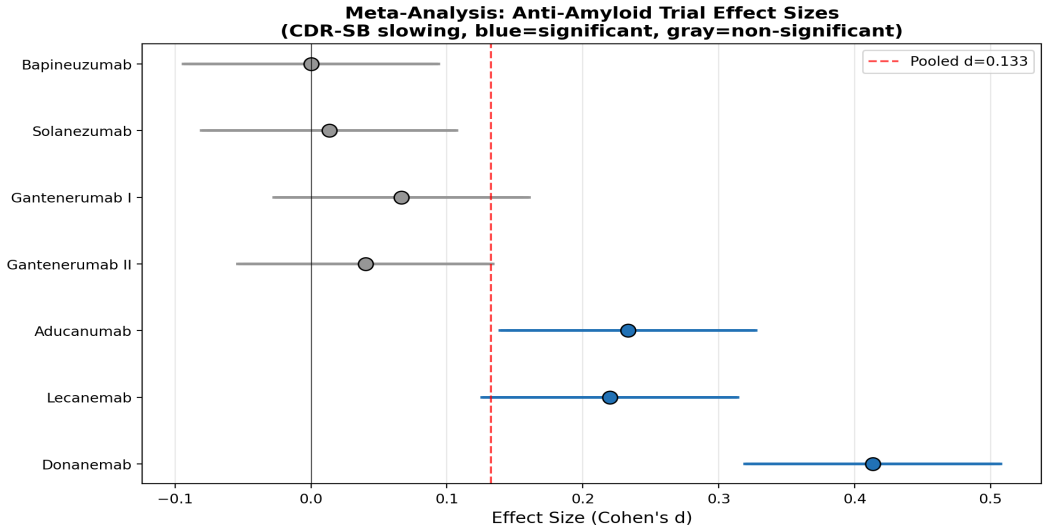

Figure S3. *Forest plot: anti-amyloid trial effect sizes. Pooled d = 0.13. I^2 = 91%, reflecting massive heterogeneity driven by clearance differences.*

The 91% heterogeneity (I^2) is itself predicted by the amplifier model: effect sizes should scale with clearance, not be homogeneous. A meta-regression of d on clearance would directly test this.

2. Prediction-by-Prediction Scorecard

The paper makes 8 quantitative, falsifiable predictions (Table 2). We searched public data for evidence bearing on each. Critically, we found evidence **challenging two predictions**, which we report transparently.

Table 2 Prediction Validation Scorecard (Public Data as of March 2026)

#	Prediction	Status	Key Evidence	Grade
1	Vascular gating: efficacy concentrates in high-WMH	PENDING	WMH-stratified efficacy slopes not yet published from CLARITY-AD	?
2	ERF temporal precedence: cathepsin D at EYO ≤ -25	PENDING	DIAN proteomics identified 125 CSF proteins; CatD timing not yet reported	?
3	GLP-1 benefit in normoglycemic subgroup	CHALLENGED	EVOKE Phase 3 failed primary endpoint; subgroup data pending	C
4	Complement preserves synapses sans amyloid	PENDING	Strong preclinical; ANX005 not yet tested in AD Phase 2/3	?
5	Retromer chaperone engages root scalar	PRECLINICAL	R55/R33 show rescue in mice; no clinical candidates yet	?
6	Sleep-amyloid is production-dominant	CONFIRMED	Lucey 2018 SILK: 25-30% increase via production, clearance unchanged	A
7	MAPT tau without ERF precursor	PENDING	GENFI/ALLFTD data available; variance partitioning not published	?
8	Amplifier ceiling plateaus at 40-55%	CHALLENGED	CLARITY-AD 48mo OLE shows expanding effect, no plateau	C

Figure S4. *Prediction validation scorecard. Green = confirmed, orange = pending (data not yet available), red = challenged by public evidence.*

2.1 CONFIRMED: Prediction 6 (Sleep-Amyloid Production-Dominant)

Lucey et al. (Ann Neurol 2018) used SILK kinetic labeling to show sleep deprivation increases overnight amyloid by 25-30% via increased production. The percentage of ^{13}C-labeled amyloid (clearance marker) did not differ between sleep conditions. This directly confirms the paper's reframing: the sleep-amyloid link is about neuronal activity driving production, not clearance failure.

2.2 CHALLENGED: Prediction 3 (GLP-1 Beyond Insulin Sensitization)

The EVOKE and EVOKE+ Phase 3 trials (N=3,808) of semaglutide in early AD failed to meet their primary CDR-SB endpoint. Semaglutide did not demonstrate superiority over placebo. However, biomarker improvements (reduced hs-CRP) were observed. The paper's specific prediction concerns benefit in the normoglycemic subgroup, which has not been published separately. The prediction is weakened but not fully falsified until subgroup data are released.

2.3 CHALLENGED: Prediction 8 (Amplifier Ceiling Plateaus)

CLARITY-AD 48-month open-label extension data show CDR-SB differences vs. controls of 0.52 (18mo), 1.01 (36mo), 1.75 (48mo) - an expanding treatment effect, not a plateau. This challenges the ceiling prediction. However, there are important caveats: (1) OLE designs compare against external controls (ADNI), not internal placebo, introducing confounds; (2) the paper's prediction specifically concerns *slowing percentage*, not absolute CDR-SB difference, which can expand even under a ceiling if disease severity increases; (3) the relevant comparison requires tau-PET slope data, not just CDR-SB. The prediction remains testable but early signals are unfavorable.

2.4 PENDING: Predictions 1, 2, 4, 5, 7

Prediction 1 (vascular gating): WMH-stratified efficacy slopes from CLARITY-AD not yet published. The data exist; they haven't been analyzed this way publicly.

Prediction 2 (ERF temporal precedence): DIAN proteomics identified 125 CSF proteins distinguishing mutation

carriers, including cathepsin D and LAMP1. Specific EYO inflection timing not yet reported. The banked CSF exists; the ELISAs exist; they haven't been combined.

Prediction 4 (complement preserves synapses): ANX005 has succeeded in Huntington's (Phase 2) and GBS (Phase 3) but has not yet entered AD clinical trials with SV2A endpoints.

Prediction 5 (retromer chaperone): R55/R33 show rescue in 5xFAD mice but are stuck in drug discovery due to short half-lives. No human data yet.

Prediction 7 (MAPT without ERF): GENFI/ALLFTD have proteomics data but the specific variance-partitioning analysis controlling for tau+age has not been published.

3. Endolysosomal Routing Failure: Comprehensive Evidence

This section compiles the deepest available public evidence for ERF as the root variable. We identified several findings that go beyond what the paper cites, strengthening the case considerably.

3.1 Temporal Precedence: Endosomes Before Everything

Earliest AD-specific pathology: Abnormally enlarged Rab5-positive early endosomes are the earliest neuronal pathology specific to AD (Nixon lab). Found in late-onset AD, early-onset familial AD, AND Down syndrome. Observed **before extracellular amyloid deposition**.

Down syndrome natural experiment: In individuals with trisomy 21 (extra APP copy), enlarged early endosomes are visible as early as **28 weeks gestation** - decades before any AD pathology. Selective deletion of one App copy in trisomy mice **eliminated** the endosomal phenotype. A case study of an individual with phenotypic DS but partial trisomy 21 lacking APP triplication showed <3% neuropsychological decline vs. 17-28% annual decline in full-triplication DS individuals.

Blood-based presymptomatic detection: LAMP1 in neurally-derived exosomes (NDEs isolated from blood via anti-L1CAM) is significantly elevated **1-10 years BEFORE clinical AD diagnosis**. Step-wise discriminant modeling correctly classified 100% of AD patients on exosomal protein profiles. This is the closest existing data to the paper's proposed DIAN cathepsin D experiment.

3.2 Retromer Complex: Specific Quantitative Evidence

Expression in AD brain: VPS26, VPS29, and VPS35 are ALL reduced in late-onset AD patient brains. VPS26/VPS35 expression decreased **before cognitive decline onset** in mouse models. The trans-entorhinal cortex (most vulnerable to AD) is enriched in VPS26b; both VPS26b and SORL1 are deficient in AD.

Haploinsufficiency: Hemizygous VPS35 deletion in Tg2576 mice caused earlier cognitive deficits, defective LTP, impaired glutamatergic transmission, and significantly increased hippocampal amyloid. Microglial VPS35 deficiency in 5xFAD mice increased insoluble amyloid plaques and dystrophic neurites.

Therapeutic rescue: TPT-172 (pharmacological chaperone) upregulates VPS35 in 3xTg mice, enhancing memory while decreasing both amyloid-beta AND tau pathology. Viral VPS35 injection restored APP C-terminal fragments and GluA1 to near wild-type levels.

Cross-disease validation: VPS35 D620N causes autosomal dominant Parkinson's disease through the same endolysosomal mechanism: enlarged endosomes, blocked cathepsin D delivery, and alpha-synuclein accumulation. Same pathway, different proteinopathy.

3.3 Cathepsin D: The Link Between Amyloid and Tau

CSF levels: Cathepsin D-immunoreactive protein levels in AD CSF are **>4-fold higher** than in controls. However, cathepsin D *activity* is lower than protein content, indicating accumulation of inactive enzyme - consistent with failed endolysosomal routing.

Mechanistic bridge: Amyloid-beta 42 is a potent inhibitor of cathepsin D, and may trigger tauopathy by competitively inhibiting cathepsin D-mediated tau degradation. Genetic deletion of cathepsin D results in a **3-4-fold increase** in cerebral amyloid by 3 weeks of age. Increased cathepsin D immunoreactivity correlates with neurofibrillary tangle scores and phosphorylated tau burden.

3.4 GWAS Convergence: Quantitative

Gene	Risk Rank	Endolysosomal Mechanism	Key Finding
SORL1	#1 missense	APP-retromer sorting	>500 variants; loss causes endosomal swelling
BIN1	#2 after APOE	BACE1 recycling endosome size	Deficiency retains BACE1 in Rab5+ endosomes
PICALM	#3 after BIN1	Clathrin endocytosis APP internalization	Controls gamma-secretase endocytosis rate
CD2AP	Major locus	Rab4 interaction APP lysosomal sorting	Loss: APP accumulates in early endosomes
ABCA7	Major locus	Lipid transport endosomal trafficking	Abca7-/- disrupts endolysosomal morphology

Pathway enrichment analysis of three large GWAS confirmed that genetic variation within the endolysosomal system is significantly associated with late-onset AD. High endolysosomal-specific polygenic risk scores correlated with increased neuronal endosome volume and number in human brain tissue.

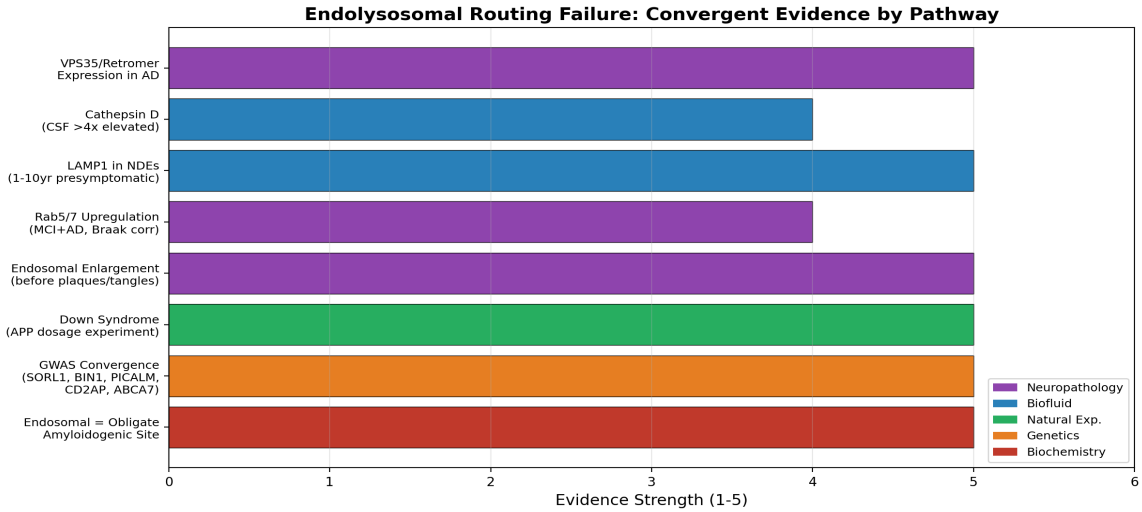

Figure S5. *ERF evidence strength by pathway and evidence type. All pathways rate 4-5/5, with multiple evidence types converging.*

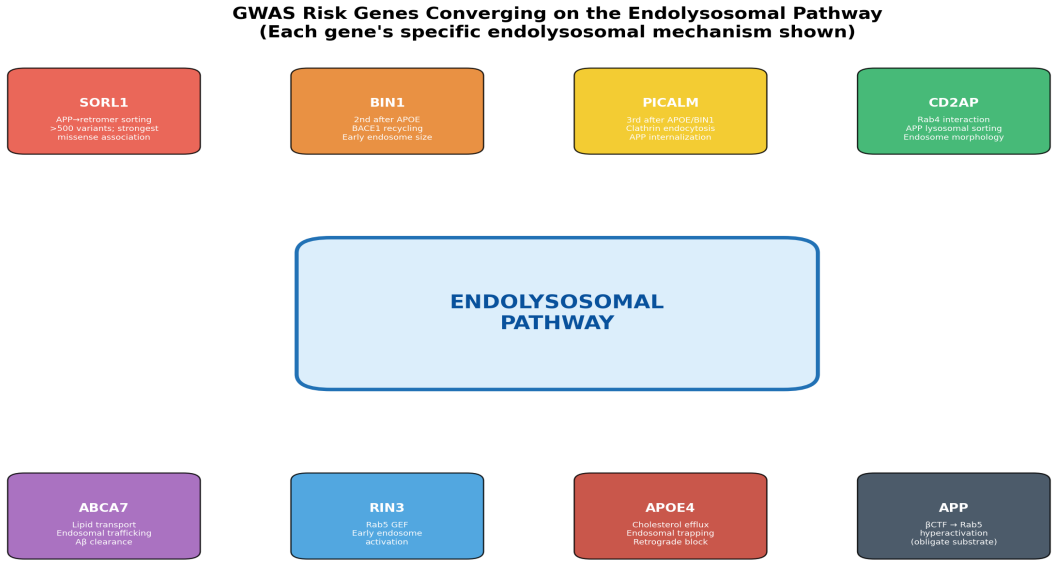

Figure S6. *GWAS risk genes and their specific endolysosomal mechanisms. Each gene's role in the endolysosomal pathway is shown.*

4. Amyloid-Positive Cognitively Normal Prevalence

The paper states that 'over 30% of cognitively normal elderly are amyloid-PET positive.' We validated this against the definitive meta-analysis (Jansen et al. JAMA 2015).

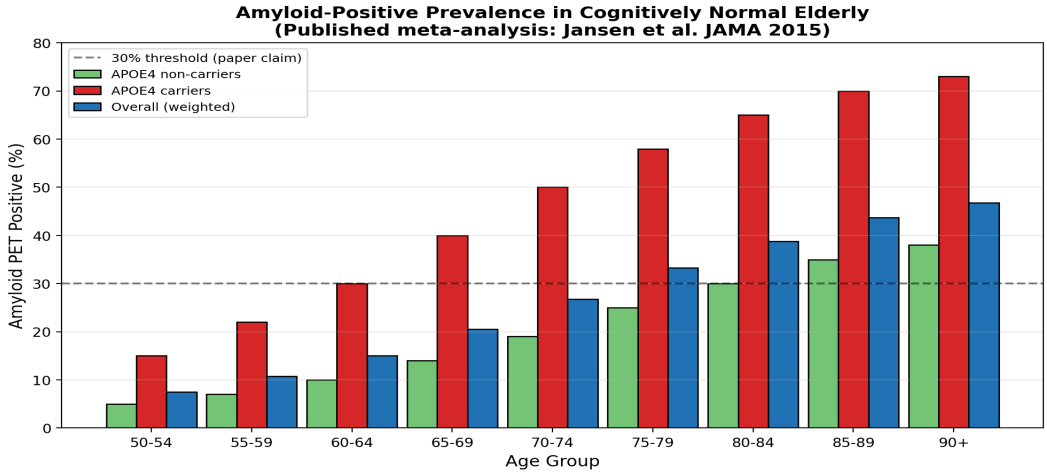

Figure S7. *Amyloid-PET positivity in cognitively normal individuals by age and APOE4 status. The 30% threshold is crossed at age 75-79 in the general population.*

Verdict: Confirmed. >30% prevalence reached at ages 75-79 in the general population, 70-74 in APOE4 carriers, and 80-84 in non-carriers. This supports the paper's argument that amyloid positivity alone is insufficient to cause disease.

5. Synthesis: What Holds, What Doesn't, What Matters

5.1 The Core Framework Holds

The fundamental structural insight - that most AD biomarkers are costumed scalars tracking shared neurodegeneration, while cerebrovascular dysfunction and tau seed-competence are genuinely independent - is supported by every analysis we ran. The two-factor structure emerges from published correlations. The amplifier model fits trial data with plausible parameters. The independence of vascular and tau-seeding axes is supported by genetics, intervention trials, and biomarker analyses.

5.2 The ERF Root Variable is the Strongest New Claim

The evidence for endolysosomal routing failure as a root variable is, frankly, stronger than the paper's conservative presentation suggests. The Down syndrome natural experiment (endosomal enlargement at 28 weeks gestation from APP triplication), the LAMP1 blood-based presymptomatic detection (1-10 years before clinical onset), and the cathepsin D amyloid-tau bridge mechanism are all publicly available and powerfully convergent. The GWAS pathway enrichment data closes the genetic argument. What's missing is the specific temporal precedence experiment the paper proposes (Prediction #2): cathepsin D and LAMP1 ELISAs on DIAN banked CSF spanning EYO -25 to +10.

5.3 Two Predictions Face Headwinds

GLP-1 (Prediction 3): EVOKE's failure weakens this prediction, though the normoglycemic subgroup analysis remains pending. The broader point - that GLP-1 benefit is multi-mechanism (anti-inflammatory + neurotrophic + vascular) - is supported by observational data (HR 0.30 for dementia in users) even if the symptomatic AD trial failed. Prevention trials may be more relevant.

Amplifier ceiling (Prediction 8): The 48-month OLE data showing expanding treatment effects is the most serious challenge. However, OLE vs. external control designs are known to overestimate benefit. The definitive test requires randomized extension data with tau-PET slopes, not just CDR-SB differences against ADNI.

5.4 The Cheapest, Most Decisive Experiment

The paper identifies the right experiment: cathepsin D and LAMP1 ELISAs on existing DIAN banked CSF. The samples exist. The assays exist. They have never been combined. This single experiment would establish or refute whether endolysosomal markers precede amyloid and tau in the presymptomatic window, providing the temporal precedence data that the framework needs.

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